Reg. No.										

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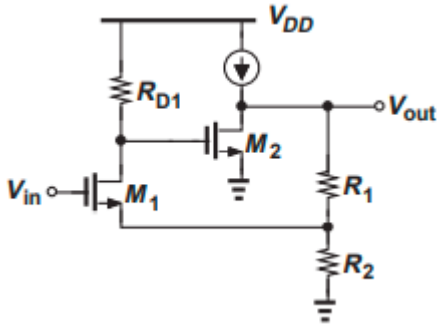
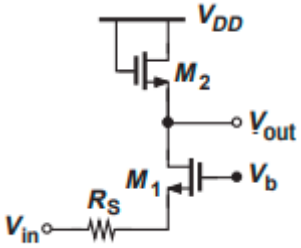
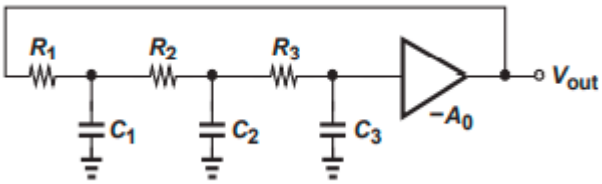
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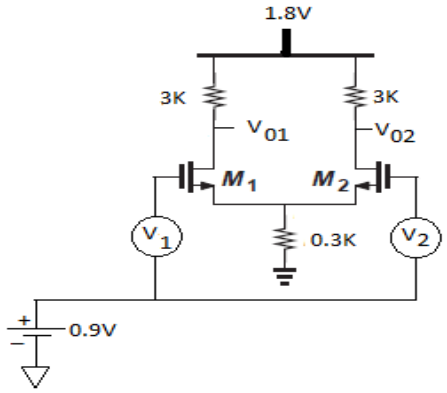
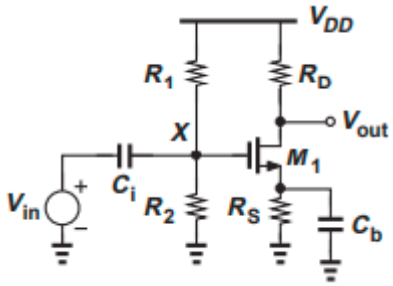
THIRD SEMESTER BTECH. (E & C) DEGREE PROCTORED ONLINE EXAMINATION
JANUARY 2022
SUBJECT: ANALOG ELECTRONICS CIRCUITS (ECE - 2151)

TIME: 75 min (9.20 AM to 10.35 AM) **MAX. MARKS: 20**

Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.
- Answer script must be a single pdf, good visibility of all texts and numbers.
- Name the file as: RegistrationNumber_SubjectCode_dd_JAN2022
- Upload correct pdf properly named well before 10.45 AM

Q. No.	Questions	M
1A	For the circuit shown in FIG.Q1A, find the Voltage gain, Input resistance and Output resistance.  FIG.Q1A	4
1B	For the circuit shown in FIG.Q1B, find the poles at high frequency. Assume λ=0.  FIG.Q1B	3
1C	For the circuit shown in FIG.Q1C, find the required voltage gain 'A_0' to get sustained oscillation and the frequency of oscillation. Assume $R_1=R_2=R_3=R$ and $C_1=C_2=C_3=C$.  FIG.Q1C	3

2A	For the circuit shown in FIG.Q2A, find the half circuit (i) Differential gain (ii) Common mode gain and (iii) CMRR. Assume the inputs V_1 and V_2 as small ac signals, 'M_1' and 'M_2' as identical MOSFETs, $\mu_n C_{ox}$ = 100μA/V^2, V_{TH} = 0.4V, (W/L) = 100 and λ=0.  FIG.Q2A	5
2B.	Design a circuit to drive a 50Ω load with a voltage gain of 0.5 and a power budget of 10mW. Assume $\mu_n C_{ox}$ = 100 μA/V^2, V_{TH} = 0.5 V, λ = 0, and V_{DD} = 1.8 V	3
2C.	For the circuit shown in FIG.Q2C, find the zeroes and poles at low frequencies for R_1=4KΩ, R_2=10KΩ, R_D=3KΩ, R_S=200Ω, C_1=0.1μF and C_b=1μF and plot the frequency response. Assume I_D =1mA, $\mu_n C_{ox}$ = 100 μA/V^2, V_{TH} = 0.5 V, (W/L) = 800 and λ = 0.  FIG.Q2C	2